European Organisation for Astronomical Research in the Southern Hemisphere

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APPLICATION FOR OBSERVING TIME

PERIOD: **100A**

Important Notice:

By submitting this proposal, the PI takes full responsibility for the content of the proposal, in particular with regard to the names of CoIs and the agreement to act according to the ESO policy and regulations, should observing time be granted.

1. Title

Category: **D–3**

Exploring the chromospheric emission and photometric variability of low-activity main-sequence stars

2. Abstract / Total Time Requested

Total Amount of Time: 0 nights VM, 31.1 hours SM

So far, the activity behaviour of main-sequence stars less active than the Sun has been poorly explored in photometry, mostly limited by the ground-based photometric precision or because of the lack of spectroscopic observations simultaneous with space-based photometry. In this proposal, we want to explore this domain of quiet stars, by performing for the first time a spectroscopic survey of 83 relatively-quiet F5 to M3 dwarfs, to detect their chromospheric emission in the Ca II lines, simultaneously with a photometric campaign with the Kepler/K2 telescope. We anticipate to unveil two populations of photometrically stable stars: a population of truly inactive stars and a population of stars where spots and faculae cancel out photometrically. This programme will have a high impact on the efficiency of exploitation of upcoming facilities like ESPRESSO (ESO), CHEOPS (ESA), and PLATO (ESA).

3.

Run	Period	Instrument	Time	Month	Moon	Seeing	Sky	Mode	Type
A	100	UVES	31.1h	any	n	n	THN	s	

4. Number of nights/hours

	Telescope(s)	Amount of time
a) already awarded to this project:		
b) still required to complete this project:		

5. Special remarks:

Since this programme targets relatively bright stars, it can be conducted under thin clouds or poor seeing conditions and, as a consequence, can be considered as a filler programme. However, because of the simultaneous observations from the K2 space mission, the observations should be performed preferentially from 2017-12-07 to 2018-02-25, or up to a few weeks before and after these dates. This period perfectly corresponds to the maximum visibility of the targets.

6. Principal Investigator:

Alexandre Santerne, alexandre.santerne@lam.fr, F, Laboratoire d'Astrophysique de Marseille (LAM)

6a. Co-investigators:

I.	Boisse	Laboratoire d'Astrophysique de Marseille (LAM),F
D.J.	Armstrong	University of Warwick,UK
T.	Lopez	Laboratoire d'Astrophysique de Marseille (LAM),F
M.	Deleuil	Laboratoire d'Astrophysique de Marseille (LAM),F

Following CoIs moved to the end of the document ...

7. Description of the proposed programme

A – Scientific Rationale: The activity of sun-like stars (caused by e.g. spots or faculae) has been studied for decades. It provides very important constraints to better understand the physical mechanisms driving the variability of the Sun. Intensive photometric and spectroscopic surveys of sun-like stars since the 80s revealed a clear correlation between the chromospheric emission, as measured by $\log R'_{HK}$ (e.g. Noyes et al., 1984, ApJ, 279, 769), and the photometric variability (see e.g. Lockwood et al., 2007, ApJS, 171, 260; Shapiro et al., 2014, A&A, 569, A38). However, this correlation has been explored so far only for relatively active stars, typically with $\log R'_{HK} \geq -4.75$ (Henry et al., 1996, AJ, 111, 439). Indeed, quiet stars, with $\log R'_{HK} \leq -4.75$, have photometric variability less than about 1mmag which is challenging to detect from the ground. As a consequence, **the activity signals of stars less active than the Sun have been poorly studied so far**.

Exoplanet-driven photometric (with e.g. the CoRoT and Kepler space missions) and spectroscopic (e.g. with the HARPS spectrograph) surveys revealed that even very quiet stars ($\log R'_{HK} \leq -4.75$) exhibit some activity signals at their rotation period, leading to challenges in the interpretation of the data. Thanks to the exquisite photometric precision of space missions like Kepler, it is possible to measure the variability of sun-like stars down to about 100ppm over several rotation periods (see e.g. McQuillan, Aigrain, & Roberts, 2012, A&A, 539, A137). However, very few of these stars have information on their chromospheric emission. The number of sun-like stars observed with both high-resolution spectroscopy (at the wavelength of Ca II H&K lines) and with space photometry is very limited. Moreover, most of these observations are not simultaneous, making interpretation difficult as activity signals might change drastically due to magnetic cycles.

In this proposal, we want to perform, for the first time, a statistically large survey of relatively quiet sun-like stars with UVES to measure their chromospheric emission in the Ca II H&K and Ca II IRT lines simultaneously with the space-based photometry from the Kepler space telescope.

Simulations of active regions (with tools like SOAP 2.0; Dumusque, Boisse & Santos, 2014, ApJ, 796, 132) show that stellar (dark) spots have a much larger photometric signature than the (bright) faculae. As a consequence, it is expected that active stars, with large photometric rotational variability, are spot-dominated. However, stars with faint photometric rotational variability might have two different behaviours: some of them could be truly quiet and the few active regions are faculae (hence they are faculae-dominated). Alternatively some could be relatively active with a mixture of both spots and faculae that compensate each other in photometry. If this is true, then we should observe **two different populations of photometrically stable stars**: the faculae-dominated stars with low $\log R'_{HK}$ values and the mixture stars (spot+faculae) with higher $\log R'_{HK}$ values (compared to the former ones). These two populations could only be observed thanks to a **simultaneous space-based photometric and high-resolution spectroscopic survey of a large sample of sun-like stars**. Note that stellar inclination might smooth out these two populations (Shapiro et al., 2016, A&A, 589, A46) but this is why the large statistical sample is critical: the distribution of stellar inclination is expected to be pure geometrical and thus could be corrected at the distribution level.

Exploring the chromospheric emission and photometric variability of low-activity main-sequence stars is very important for 2 main reasons: 1- this programme will provide insights to understand the activity of stars less active than the Sun, in particular to see how the activity behaviour changes with regard to the spectral type and with the size of the convective layer ; 2- stellar activity is currently the main limitation to exoplanet-search programmes with spectrographs like HARPS. To fully exploit the data from the new generation spectrograph ESPRESSO@VLT, it is extremely important to better understand the activity signals of relatively quiet stars that will be prime targets for this facility. This will be even more important in the context of the PLATO / ESA space mission as important decisions regarding the expensive follow-up observations will have to be taken, mostly based on the high-precision light curve and the expected activity level of the stars. Our team is responsible for the ranking of the PLATO exoplanets for the follow-up observations, hence this programme will allow us to optimise the strategy to maximise the number of Earth-like planets in the habitable zone that will be detected and characterised. It will also provides estimates of the photometric variability of CHEOPS (ESA) targets, based on their observed $\log R'_{HK}$ value. **This programme, which can be considered as filler, will therefore have a high impact on both stellar physics and on the exploitation of next generation instruments like ESPRESSO, CHEOPS, and PLATO.**

Such a simultaneous photometry and spectroscopy campaign is only possible because the Kepler telescope will observe during the K2 campaign C16 in forward-facing mode, i.e. pointing towards the Earth, which will allow large simultaneous ground- and space-based observations. This is not possible during normal K2 campaigns (in rear-facing mode). Over the 3 years lifetime of the K2 mission, only the campaigns C9 and C16 are done in the forward-facing mode (note that C9 was dedicated to the galactic center for microlensing detections). The K2 campaign C16 (in operation from 2017 Dec. 7 to 2018 Feb. 25) is therefore **a unique opportunity to perform such a simultaneous photometry and spectroscopy campaign**. Moreover, the field-of-view of C16 will overlap the one of C5. As a consequence, more than 3000 stars that will be observed early 2018 were already observed by K2 mid-2015 (campaign C5), allowing us **to select the best relatively-quiet sun-like stars** for this campaign.

Such a programme was not possible on a large sample of the *CoRoT* or *Kepler* stars since most of them were too faint to measure precise $\log R'_{HK}$ values or the spectrographs used do not cover the Ca II H&K lines. Only

7. Description of the proposed programme and attachments

Description of the proposed programme (continued)

Bastien et al. (2014, AJ, 147, 29) attempted such simultaneous photometric and spectroscopic campaign but only on a dozen stars, including a substantial fraction of subgiants. Salabert et al., (2016, A&A, 596, A31) also performed a spectroscopic survey of 18 solar analogs observed by Kepler, but the spectroscopic data were collected 2 years after the Kepler data. So far, there is no report in the literature of a simultaneous photometric and spectroscopic surveys of a large sample of main-sequence solar-type stars.

B – Immediate Objective: In this proposal, we want to measure the chromospheric emission in the Ca II H&K and Ca II IRT lines of a sample of 83 bright solar-type stars (from F5 to M3) already observed during the *K2* campaign C5 (mid-2015) and that will be re-observed during campaign C16 from 2017-12-07 to 2018-02-25 (dates secured within a few days). 75% of these stars exhibited in mid-2015 a photometric variability lower than 1mmag and should be considered as relatively quiet stars. The spectroscopic observations we are requesting here should be preferentially simultaneous with the K2 observations to avoid magnetic cycles affecting the results. However, if the spectroscopic observations are taken a few weeks before or after the K2 photometry, we do not expect magnetic cycles to significantly affect our conclusions. From the emission in the Ca II H&K line, we will derive the $\log R'_{HK}$ following the procedure described in Gomes da Silva et al. (2011, A&A, 534, A30). This is the most important value for this study as it is widely used in the literature. We are also interested in the Ca II IRT lines to prepare the upcoming near-infrared exoplanet spectroscopic surveys (with e.g. SPIRou@CFHT, NIRPS@ESO, CRIRES@VLT) and for possible comparison with the RAVE and GAIA RVS data.
To measure precisely the chromospheric emission in the Ca II H&K and Ca II IRT lines, **we need one high-resolution (R≥40 000) and high-SNR (∼50) spectrum covering the corresponding wavelength domains**. This should provide $\log R'_{HK}$ values with a precision at the level of $\approx 1\%$, based on our experience with HARPS. Note that the $\log R'_{HK}$ should vary with the rotational period, but since these stars are relatively quiet, the variation is expected to be low, if not within the photon noise.
Our sample selection is described in the Target notes. We have also included a few active stars that exhibit photometric variability up to 2% (see Fig. 1). These stars will be used as a comparison sample.
Note that this programme can not be performed using the GAIA data since (1) GAIA's observing cadence will not be as good as the 80d-long 30min cadence of the K2 photometry and (2) GAIA RVs data does not include the Ca II H&K lines which are the better lines to monitor chromospheric emission compared to the IRT lines. However, GAIA RVS and RAVE data can perfectly complement this sample, to tentatively explore magnetic cycles on low-activity stars.

Attachments (Figures)

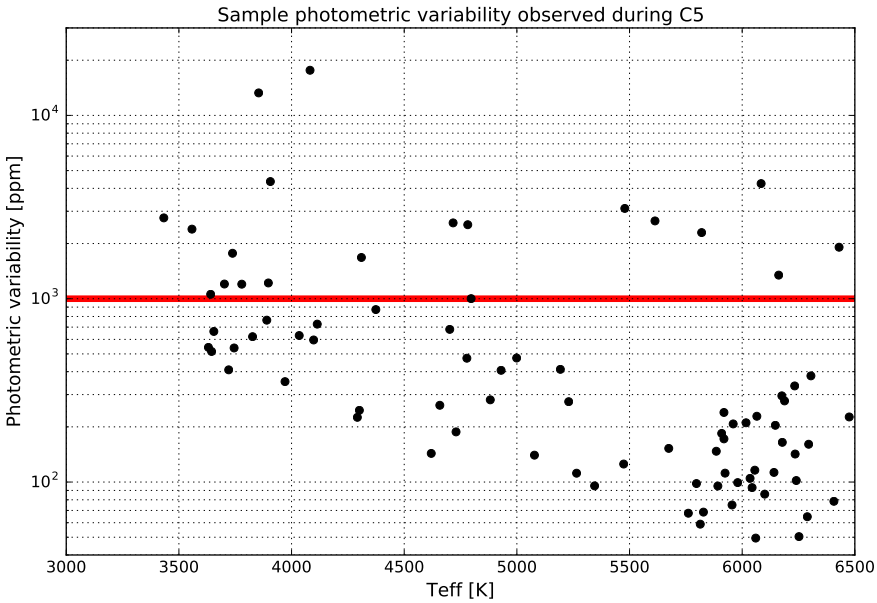

Fig. 1: Photometric variability over 80 days of the sample stars as observed during the K2 campaign C5 (mid 2015) as function of the effective temperature of the star. The red line indicates the 1mmag floor of ground-based photometric accuracy. The photometric variability of 63 targets out of the 83 (hence 75%) can not be detected from the ground.

8. Justification of requested observing time and observing conditions

Lunar Phase Justification: The target stars are relatively bright. They can be observed at any Lunar phase. A minimum Moon distance of 30°should still be used for the observations.

Time Justification: (including seeing overhead) We used the ESO ETC to determine the exposure time. We adjusted the exposure time in order to have a S/N of 50 in the continuum near the Ca II H&K lines for FGK dwarfs. For the M dwarfs, for which the photospheric emission is faint compared to the chromospheric emission (even for quiet stars), we adjusted the exposure time to have a S/N of 15 in the continuum near the Ca II H&K lines. This gives a S/N of about 150 in the continuum close to the Ca II IRT lines for all stars. We used overheads of 580s. This gives 31.1h of UVES time for 1 spectrum on each target.

8a. Telescope Justification:

To complete this programme, we need high-resolution and high-SNR spectra of the Ca II H&K and IRT lines for 83 stars with a magnitude ranging from 7.7 to 13 in the R band (median at 9.9). Only UVES@VLT allows efficient observation of these two spectral domains, with high-resolution in one observation. The Ca II IRT line is not in the wavelength domain of HARPS. X-SHOOTER is of too low spectral resolution to derive precise $\log R'_{HK}$ values.

We will use the UVES blue arm with a 1″ slit for a resolution of 40 000 in the blue. We request for the DIC#2 to observe with both the blue and red arms. The red arm, which comes free with no additional telescope time, will allow us to measure the Ca II IRT line. We did not request for the image slicer as it would impact quite a lot the observations if they are taken at relatively high airmass, which is likely given the high declination of the stars. The targets are close to the ecliptic plane, and are therefore well observable from Paranal, between 2 and 5 hours every nights over the K2 campaign period.

8b. Observing Mode Justification (visitor or service):

Service mode is preferred for this programme on UVES.

8c. Calibration Request:

Standard Calibration

9. Report on the use of ESO facilities during the last 2 years

Large Programmes on HARPS 198.C-0169 (PI: Santerne), 196.C-0042 (PI: Santos) to search and characterise low-mass planets around solar-type stars. With only a few months of data, we are preparing several planet-discovery papers, including a planet of about 3 Mearth. However, the activity of stars is a limitation to either fully exploit our data or to reach an exquisite precision on the planet fundamental parameters. Therefore this timely programme will allow us to better exploit our large programme data.
We already used UVES to get, for the first time, radial velocity observations of a microlensing detection, which led to 2 letters published in A&A (Boisse et al., 2015, A&A, 582, L11 ; Santerne et al., 2016, A&A, 595, L11).

9a. ESO Archive - Are the data requested by this proposal in the ESO Archive (http://archive.eso.org)? If so, explain the need for new data.

We request data simultaneous with a foreseen K2 campaign. None of the observations already in the archive, if any, is simultaneous.

9b. GTO/Public Survey Duplications:

These stars are not part of GTO or public surveys.

10. Applicant's publications related to the subject of this application during the last 2 years

– **Armstrong, D. J.**, Kirk, J., Lam, K. W. F., et al. 2015, A&A, 579, A19: "K2 Variable Catalogue: Variable stars and eclipsing binaries in K2 campaigns 1 and 0"

– **Barros, S. C. C.**, Almenara, J. M., **Deleuil, M.** et al. 2014, A&A, 569, A74: "Revisiting the transits of CoRoT-7b at a lower activity level"

– Bruno, G., **Deleuil, M.**, Almenara, J.-M., et al. 2016, A&A, 595, A89: "Disentangling planetary and stellar activity features in the CoRoT-2 light curve"

– **Dumusque, X.**, **Boisse, I.**, & **Santos, N. C.** 2014, ApJ, 796, 132: "SOAP 2.0: A Tool to Estimate the Photometric and Radial Velocity Variations Induced by Stellar Spots and Plages"

– Gomes da Silva, J., **Santos, N. C.**, **Boisse, I.**, **Dumusque, X.**, & Lovis, C. 2014, A&A, 566, A66: "On the long-term correlation between the flux in the Ca II H & K and Hα lines for FGK stars"

– Mayor, M., Lovis, C., & **Santos, N. C.** 2014, Nature, 513, 328: "Doppler spectroscopy as a path to the detection of Earth-like planets"

– Oshagh, M., **Santos, N. C.**, **Boisse, I.**, et al. 2013, A&A, 556, A19: "Effect of stellar spots on high-precision transit light-curve"

– **Santos, N. C.**, Mortier, A., Faria, J. P., et al. 2014, A&A, 566, A35: "The HARPS search for southern extra-solar planets. XXXV. The interesting case of HD 41248: stellar activity, no planets?"

– **Santerne et al.**, in prep.: "Precise planetary parameters from the ESO-K2 large programme. I. K2-XX b: a 3 Mearth rocky planet transiting a very active K dwarf"

11. List of targets proposed in this programme

Run	Target/Field	α(J2000)	δ(J2000)	ToT	Mag.	Diam.	Additional info	Reference star
A	EPIC211814413	08 40 33.940	+17 31 14.94	0.9	12.91	0	Teff=3433K,PhotVar=2761ppm	
A	EPIC211818581	08 56 56.604	+17 34 46.43	1.0	12.97	0	Teff=3558K,PhotVar=2396ppm	
A	EPIC211845940	08 43 10.673	+17 57 44.20	0.5	11.87	0	Teff=3631K,PhotVar=543ppm	
A	EPIC211892034	09 05 42.968	+18 36 33.75	0.4	11.58	0	Teff=3641K,PhotVar=1057ppm	
A	EPIC211765962	08 49 52.168	+16 49 59.22	0.8	12.64	0	Teff=3645K,PhotVar=516ppm	
A	EPIC211480655	09 06 14.044	+12 51 34.20	0.6	12.21	0	Teff=3655K,PhotVar=663ppm	
A	EPIC211699606	08 41 30.097	+15 54 37.07	1.0	12.97	0	Teff=3702K,PhotVar=1203ppm	
A	EPIC212002676	08 50 36.916	+20 17 54.99	0.9	12.86	0	Teff=3721K,PhotVar=409ppm	
A	EPIC211945363	08 57 15.421	+19 24 17.66	0.6	12.35	0	Teff=3738K,PhotVar=1770ppm	
A	EPIC212006344	08 25 54.314	+20 21 34.44	0.7	12.47	0	Teff=3745K,PhotVar=539ppm	
A	EPIC211698189	08 42 19.291	+15 53 28.70	0.9	12.83	0	Teff=3779K,PhotVar=1200ppm	
A	EPIC211529603	08 42 26.207	+13 33 25.85	0.9	12.83	0	Teff=3827K,PhotVar=621ppm	
A	EPIC212162615	08 41 42.888	+23 18 46.77	0.7	12.61	0	Teff=3854K,PhotVar=13266ppm	
A	EPIC212118494	08 54 01.083	+22 23 12.00	0.7	12.62	0	Teff=3890K,PhotVar=764ppm	
A	EPIC211401683	08 50 34.387	+11 39 56.68	0.8	12.65	0	Teff=3897K,PhotVar=1219ppm	
A	EPIC212032390	08 27 49.522	+20 47 40.87	0.7	12.51	0	Teff=3906K,PhotVar=4357ppm	
A	EPIC212092574	08 28 44.924	+21 52 42.93	0.9	12.86	0	Teff=3971K,PhotVar=353ppm	
A	EPIC211481688	09 07 38.573	+12 52 29.33	0.5	11.81	0	Teff=4034K,PhotVar=630ppm	
A	EPIC211589755	09 01 46.289	+14 24 01.32	0.3	10.68	0	Teff=4082K,PhotVar=17630ppm	
A	EPIC211427097	08 52 10.192	+12 02 53.67	0.5	11.72	0	Teff=4098K,PhotVar=596ppm	
A	EPIC212096658	08 27 00.897	+21 57 24.69	0.2	10.21	0	Teff=4114K,PhotVar=727ppm	
A	EPIC211527014	08 45 03.812	+13 31 14.90	0.5	11.86	0	Teff=4292K,PhotVar=225ppm	
A	EPIC211418016	08 39 03.847	+11 54 42.86	0.5	11.74	0	Teff=4301K,PhotVar=246ppm	
A	EPIC211695537	08 40 20.617	+15 51 21.73	0.5	11.61	0	Teff=4310K,PhotVar=1679ppm	
A	EPIC211531076	08 37 37.368	+13 34 41.50	0.5	11.91	0	Teff=4374K,PhotVar=873ppm	
A	EPIC212024392	08 54 40.270	+20 39 34.85	0.3	11.06	0	Teff=4620K,PhotVar=143ppm	
A	EPIC212037002	08 25 45.678	+20 52 17.25	0.5	11.78	0	Teff=4658K,PhotVar=262ppm	
A	EPIC211827122	08 41 48.972	+17 42 06.25	0.4	11.56	0	Teff=4702K,PhotVar=680ppm	
A	EPIC211720173	08 42 52.822	+16 11 15.61	0.4	11.27	0	Teff=4717K,PhotVar=2592ppm	
A	EPIC212093295	08 26 44.042	+21 53 33.30	0.4	11.50	0	Teff=4730K,PhotVar=188ppm	
A	EPIC211545368	09 07 20.323	+13 46 39.49	0.4	11.28	0	Teff=4778K,PhotVar=474ppm	
A	EPIC211622479	09 04 25.407	+14 50 58.75	0.4	11.25	0	Teff=4782K,PhotVar=2534ppm	
A	EPIC211949026	08 44 42.740	+19 27 30.21	0.3	10.81	0	Teff=4797K,PhotVar=1001ppm	
A	EPIC211487667	08 39 57.666	+12 57 39.15	0.5	11.90	0	Teff=4882K,PhotVar=281ppm	
A	EPIC211698705	08 44 42.801	+15 53 53.16	0.4	11.47	0	Teff=4930K,PhotVar=407ppm	

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Target Notes: We selected our sample stars in the overlap between the C5 and C16 target list of the K2 mission. We then selected all dwarf stars having a Teff < 6500K. We then kept the stars with magnitude in the Kepler bandpass (similar to R band) brighter than 13 for Teff < 4000K, brighter than 12 for 4000K <Teff < 5000K, and brighter than 10 for the other ones. We removed from this sample eclipsing binaries and exoplanet hosts. This gave us a sample of 83 stars. We reduced their K2 light curve from campaign C5 and computed their photometric variability over the 80 days of the campaign (see Fig. 1).

12. Scheduling requirements

13. Instrument configuration

Period	Instrument	Run ID	Parameter	Value or list
100	UVES	A	DIC-2	Standard setting: 390+760

6b. Co-investigators:

...continued from Box 6a.

M.	Hobson	Laboratoire d'Astrophysique de Marseille (LAM),F
S.	Barros	IA - Instituto de Astrofisica e Ciencias do Espaco,P
X.	Dumusque	Observatoire Astronomique de l'Universite de Geneve,CH
N.C.	Santos	IA - Instituto de Astrofisica e Ciencias do Espaco,P
M.	Lendl	Space Research Institute (IWF),Austrian Academy of Sciences,AT

11a. List of targets proposed in this programme

Run	Target/Field	α(J2000)	δ(J2000)	ToT	Mag.	Diam.	Additional info	Reference star
	...continued from box 11.							
A	EPIC211696686	08 39 52.174	+15 52 17.40	0.5	11.69	0	Teff=4999K,PhotVar=475ppm	
A	EPIC211529519	08 40 33.503	+13 33 23.00	0.2	9.91	0	Teff=5078K,PhotVar=140ppm	
A	EPIC212164712	08 57 03.700	+23 21 47.27	0.2	9.38	0	Teff=5193K,PhotVar=412ppm	
A	EPIC211529184	08 42 26.739	+13 33 05.83	0.2	9.84	0	Teff=5230K,PhotVar=274ppm	
A	EPIC212004790	09 06 45.784	+20 20 04.95	0.2	9.61	0	Teff=5265K,PhotVar=112ppm	
A	EPIC212118357	08 52 35.602	+22 23 03.35	0.2	9.54	0	Teff=5345K,PhotVar=95ppm	
A	EPIC212166808	08 48 28.919	+23 24 44.22	0.2	9.32	0	Teff=5474K,PhotVar=125ppm	
A	EPIC211750200	08 54 41.509	+16 36 40.35	0.2	8.45	0	Teff=5479K,PhotVar=3108ppm	
A	EPIC211693508	08 39 21.418	+15 49 42.87	0.2	9.84	0	Teff=5613K,PhotVar=2657ppm	
A	EPIC212110594	08 28 19.429	+22 13 48.76	0.2	9.93	0	Teff=5674K,PhotVar=153ppm	
A	EPIC212030997	08 31 02.749	+20 46 18.15	0.2	9.99	0	Teff=5761K,PhotVar=68ppm	
A	EPIC211494107	09 06 35.268	+13 03 20.59	0.2	9.89	0	Teff=5797K,PhotVar=98ppm	
A	EPIC211695331	08 38 08.464	+15 51 11.83	0.2	9.64	0	Teff=5814K,PhotVar=59ppm	
A	EPIC211998546	08 54 00.504	+20 13 49.42	0.2	8.99	0	Teff=5820K,PhotVar=2295ppm	
A	EPIC211818901	08 45 08.820	+17 35 04.67	0.2	9.78	0	Teff=5828K,PhotVar=69ppm	
A	EPIC211999267	09 06 54.555	+20 14 31.81	0.2	9.80	0	Teff=5885K,PhotVar=147ppm	
A	EPIC211842729	08 52 22.137	+17 55 07.74	0.2	9.66	0	Teff=5892K,PhotVar=95ppm	
A	EPIC211748624	08 51 12.757	+16 35 21.85	0.2	9.66	0	Teff=5909K,PhotVar=184ppm	
A	EPIC211505372	09 04 15.681	+13 12 54.01	0.2	9.68	0	Teff=5919K,PhotVar=240ppm	
A	EPIC212119741	08 32 32.537	+22 24 40.14	0.2	8.99	0	Teff=5919K,PhotVar=172ppm	
A	EPIC211811652	08 41 21.819	+17 28 55.49	0.2	9.83	0	Teff=5924K,PhotVar=112ppm	
A	EPIC211838719	08 55 26.681	+17 51 53.71	0.2	9.55	0	Teff=5955K,PhotVar=75ppm	
A	EPIC212121350	08 29 53.627	+22 26 32.26	0.2	9.87	0	Teff=5960K,PhotVar=208ppm	
A	EPIC211677447	08 54 33.522	+15 36 36.44	0.2	9.75	0	Teff=5980K,PhotVar=99ppm	
A	EPIC212130717	08 30 22.680	+22 37 58.16	0.2	9.88	0	Teff=6017K,PhotVar=211ppm	
A	EPIC211754103	08 54 23.068	+16 39 55.87	0.2	9.52	0	Teff=6035K,PhotVar=105ppm	
A	EPIC211687013	08 56 53.097	+15 44 23.91	0.2	10.00	0	Teff=6044K,PhotVar=93ppm	
A	EPIC212010530	08 29 27.246	+20 25 47.11	0.2	8.79	0	Teff=6056K,PhotVar=116ppm	
A	EPIC211936772	08 50 14.658	+19 16 34.37	0.2	8.86	0	Teff=6060K,PhotVar=49ppm	
A	EPIC211946900	08 44 47.394	+19 25 37.67	0.2	9.21	0	Teff=6065K,PhotVar=228ppm	
A	EPIC211947631	08 43 05.939	+19 26 15.24	0.2	9.84	0	Teff=6084K,PhotVar=4247ppm	
A	EPIC211499698	08 40 54.092	+13 08 10.28	0.2	9.03	0	Teff=6100K,PhotVar=86ppm	
A	EPIC211497202	08 44 43.584	+13 06 00.84	0.2	9.43	0	Teff=6141K,PhotVar=113ppm	
A	EPIC211825485	08 54 01.439	+17 40 42.65	0.2	8.85	0	Teff=6147K,PhotVar=204ppm	
A	EPIC211936988	09 05 24.758	+19 16 45.33	0.2	9.36	0	Teff=6162K,PhotVar=1345ppm	
A	EPIC211841501	08 44 11.960	+17 54 07.87	0.2	9.95	0	Teff=6176K,PhotVar=295ppm	
A	EPIC211950227	08 40 25.542	+19 28 32.77	0.2	9.82	0	Teff=6178K,PhotVar=165ppm	
A	EPIC211707875	08 53 52.394	+16 01 10.21	0.2	9.64	0	Teff=6188K,PhotVar=277ppm	
A	EPIC212013281	08 57 47.749	+20 28 28.16	0.2	9.43	0	Teff=6233K,PhotVar=334ppm	
A	EPIC211516730	08 42 18.735	+13 22 28.90	0.2	9.09	0	Teff=6235K,PhotVar=142ppm	
A	EPIC212109644	08 54 18.706	+22 12 40.54	0.2	7.74	0	Teff=6240K,PhotVar=102ppm	
A	EPIC211811359	08 39 31.506	+17 28 41.52	0.2	8.47	0	Teff=6252K,PhotVar=50ppm	
A	EPIC211725118	08 38 10.716	+16 15 18.82	0.2	9.47	0	Teff=6289K,PhotVar=65ppm	
A	EPIC211706914	08 55 31.018	+16 00 24.05	0.2	9.62	0	Teff=6295K,PhotVar=161ppm	
A	EPIC211954550	08 41 26.978	+19 32 32.89	0.2	9.80	0	Teff=6305K,PhotVar=380ppm	
A	EPIC212086571	08 31 47.893	+21 45 45.02	0.2	9.84	0	Teff=6407K,PhotVar=79ppm	
A	EPIC211922975	08 43 07.053	+19 04 06.08	0.2	9.44	0	Teff=6430K,PhotVar=1910ppm	
A	EPIC211947064	09 06 28.260	+19 25 45.33	0.2	9.64	0	Teff=6475K,PhotVar=227ppm	